

# Yukta Jain

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## EXPERIENCE

### Experiqs Pvt Ltd

Apr 2026 – Present

#### Robotics Intern

Mumbai

- Implemented a vision-guided pick-and-place robotic system on a **RoArm-M2-S** arm using **Grounding DINO** (Vision-Language Model) for real-time perception and **ROS2** for sensor-to-actuator communication
- Derived forward and inverse kinematics to convert camera-frame object coordinates into joint-space commands using **Denavit-Hartenberg** convention
- Implemented **cubic polynomial** trajectory to produce smooth motion profiles for the manipulator

### L&T Finance

Jul 2024 – Sept 2025

#### Graduate Engineer Trainee

Mumbai

- Implemented automation framework utilizing **Selenium** and **Java**, reducing manual efforts by **40%**

### TIH-IOT CHANAKYA Fellowship Program, IITB

Mar 2023 – Jan 2024

#### Student Fellow

Mumbai

- Developed an **IoT-enabled EMG-controlled wheelchair**, by processing EMG (electromyography) signals using Muscle Sensor V3.0 and ESP32 to enable muscle-based assistive control
- Integrated IoT-based battery and stability monitoring for real-time notifications using **HC-SR04** and **voltage** sensor

## PROJECTS

### Line Follower Robot

[Link To Repo](#)

- Developed baremetal firmware of Nucleo-F103rb for an autonomous line follower robot

### STM32F-103 Bootloader

[Link To Repo](#)

- Implemented a bootloader using libopenm3 in Linux handling vector table relocation and application jumping

### NucleoF103RB driver library

[Link To Repo](#)

- Implemented driver files for accessing, reading and transmitting data to Nucleo-F103rb via different peripherals i.e. GPIOs, SPI, I2C, UART, etc.
- Implemented I2C driver functions to verify, configure and read the registers of the MPU6050 with Nucleo-F103rb

### Mybot

[Link To Repo](#)

- Built a differential drive robot in ROS2 Jazzy with URDF, simulated in gz sim, visualized in Rviz2 and enabled autonomous navigation using the Nav2 stack and SLAM Toolbox for real-time mapping and localization

### ROS2 Fundamental Examples

[Link To Repo](#)

- Implemented publisher subscriber nodes in ROS2 Jazzy to read HC-SR04 sensor data, process obstacle detection logic, and send motion control commands back to the Arduino nano via pyserial Library

## SKILLS

**Languages\Frameworks** : ROS2, Python, C, C++

**Development Boards** : RaspberryPi 5, Jetson Orin Nano Super, STM32F103rb, ESP32, Arduino Uno/Due

**Hardware Tools** : Logic analyzer, Multimeter, Oscilloscope

**Software Tools** : MATLAB, Linux(Ubuntu), STM32 Cube IDE, Arduino IDE, VS Code

## EXTRACURRICULAR ACTIVITIES

### Electronics Team Member - DJS Robocon

Aug 2022-Dec 2023

- Interfaced sensors, motors and actuator (DCV valve), designed and soldered the perfboard circuit and facilitated comprehensive testing for robot mechanisms

## EDUCATION

### Dwarkadas J. Sanghvi College of Engineering

Bachelor Of Electronics Engineering

CGPA-8.60

## CERTIFICATIONS

Mastering Microcontroller And Embedded Driver Development

[Link To Certificate](#)